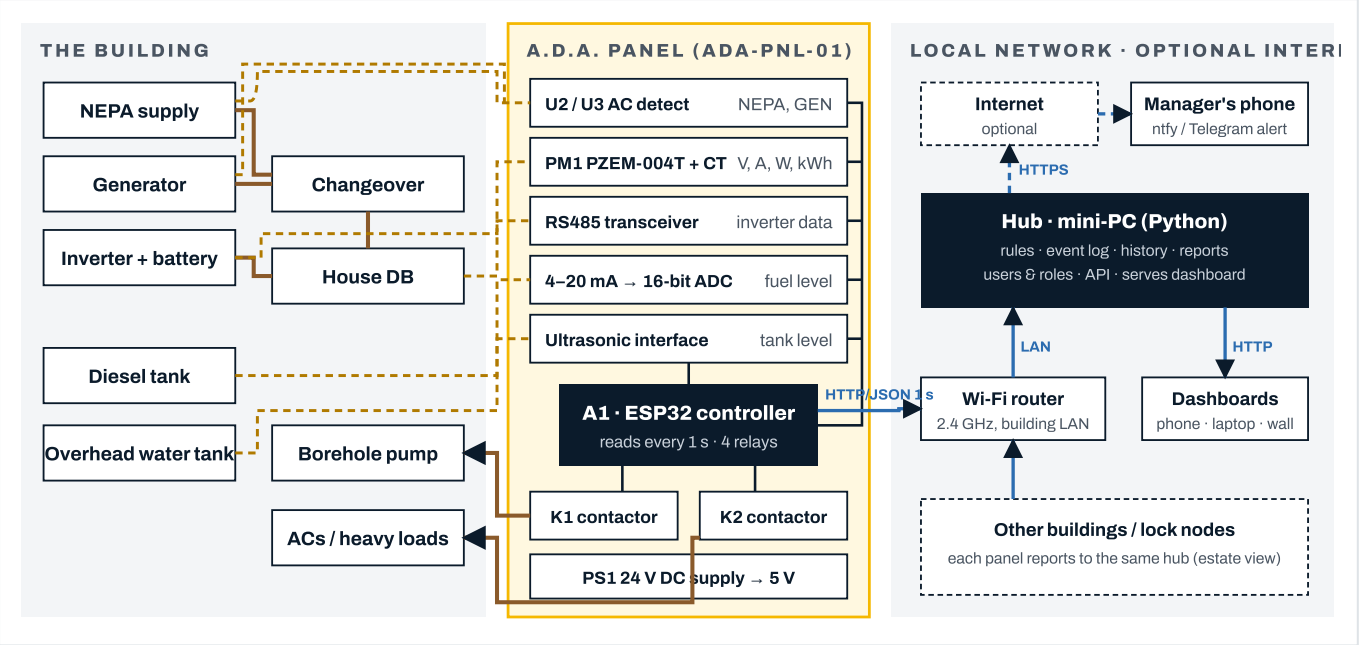
System architecture & specification

How A.D.A. is connected, what each device measures, and how the data reaches the dashboard and the manager's phone.

1 The system at a glance



230 V power · sensing (isolated) · low-voltage signal inside the panel · network (Wi-Fi / LAN) · internet, optional (alerts only)

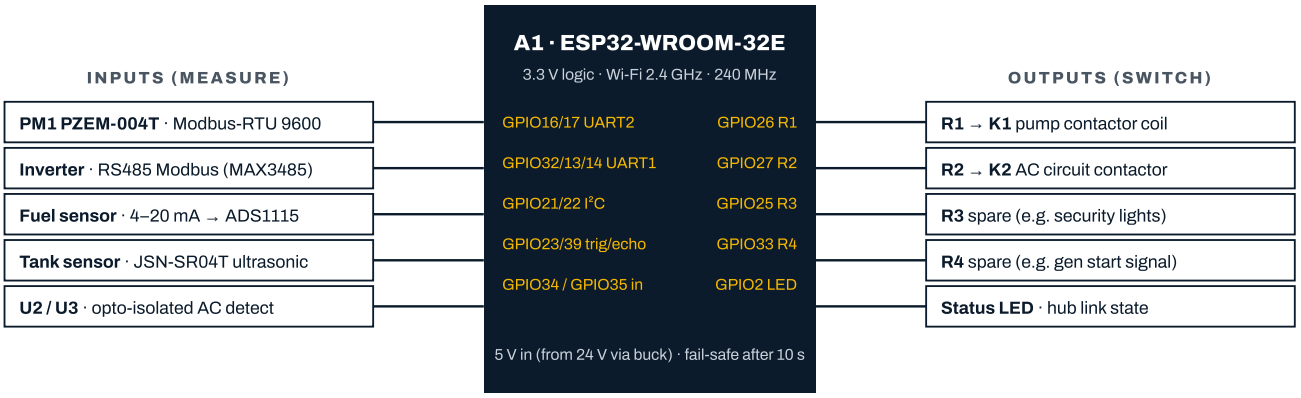
Three layers

- **Measure**: isolated sensors on the supply, the inverter, the diesel tank and the water tank.
- **Decide**: the hub runs the rules and keeps the log; the controller switches contactors.
- **Show & alert**: the dashboard on any phone or laptop on the building network; alerts to the manager's phone.

Design principles

- **Local first**: everything runs in the building. Internet is only used to push alerts.
- **Fail-safe**: if the controller loses the hub for 10 s, relays drop to safe states and the building runs as it does today.
- **Mains is isolated**: no 230 V reaches the controller; all mains work by a qualified electrician.

2 The controller and how each device connects



Every input reaches the ESP32 at low voltage through an isolating module: optocouplers for AC presence, the PZEM's isolated interface for power, an RS485 transceiver for the inverter, a 4–20 mA current loop for the fuel probe (robust over long cable runs from the generator house) and a level-shifted echo line for the ultrasonic tank sensor. Outputs drive relay coils, and the relays drive contactor coils: the controller never carries load current.

3 Device and sensor specifications

DEVICE	TYPE	MEASURES / DOES	INTERFACE	RANGE / ACCURACY	POWER	LOCATION
A1 Controller	ESP32-WROOM-32E on DIN carrier, 4 opto-isolated relays	Reads every sensor, runs fail-safe, reports to hub	`Wi-Fi 2.4 GHz; HTTP/JSON`	240 MHz dual-core; 520 KB RAM	5 V, ≈ 1 W	Panel, rail 2
PS1 Power supply	DIN-rail 230 V AC → 24 V DC, 30 W, + buck to 5 V	Powers controller, relays, sensors	–	Isolated, short-circuit protected	Input 85–264 V AC	Panel, rail 1
PM1 Power meter	PZEM-004T v3.0 with 100 A split-core CT	Voltage, current, power, energy, frequency, power factor of the house supply	`UART Modbus-RTU, 9600 baud`	V 80–260 V ±0.5%; A 0–100 A ±0.5%	Self-powered from mains side	Panel; CT on house live
U2 / U3 Source detect	Opto-isolated 230 V AC presence modules	Is NEPA present? Is the generator running? (before the changeover)	`Digital input, isolated`	Response < 50 ms	From mains side	Panel, rail 1
Inverter link	MAX3485 RS485 transceiver (or shunt + INA226 where no port)	Battery %, battery V and A, output load, inverter mode	`RS485 Modbus-RTU (brand register map) / I²C`	As reported by the inverter; shunt ±0.1%	3.3 V	Panel → inverter data port
Fuel level sensor	Capacitive diesel probe, cut to tank depth, 4–20 mA	Litres in the generator tank; refills, use, drops while gen off	`4–20 mA → 165 Ω → ADS1115 16-bit ADC (I²C)`	±1% of full scale; ≈ ±0.5 L on a 50 L tank	24 V loop	Generator tank lid
Tank level sensor	JSN-SR04T waterproof ultrasonic	Water level in the overhead tank (%) for pump control	`Trigger / echo GPIO (echo level-shifted)`	Range 25–450 cm, ±1 cm	5 V	Tank lid, cable to panel
K1 / K2 Contactors	Modular 2-pole, 25 A AC-1, 24 V coil	Switch the borehole pump and the AC circuit	`Coil driven by relay R1 / R2`	Rated for motor and AC loads; size to the actual pump	Coil ≈ 2–3 W	Panel, rail 2
Hub	Small always-on mini-PC (or	Rules, event log, history, alerts, users	`LAN HTTP; outbound HTTPS`	Python 3 standard library, no cloud	10–15 W	Office or panel room, on

4 How the data flows



Every second

- The controller samples each sensor, filters noise (e.g. averages the fuel probe over 30 s so sloshing is not read as theft) and sends one JSON report to the hub: `{"source":"grid","readings":{"fuel_l":31.5,"battery":64,"tank":48,"voltage":228},"relays":{"pump":false}}`
- The hub runs the rules and replies with the relay states the controller must apply.
- Dashboards poll the hub every second; if a reading is older than 5 s they say **Not live** and refuse to switch.

On events

- Each change (source, switch, spike, refill, drop, delivery, offline) is written to the event log with its time and reason.
- Alerts (diesel drop, short delivery, low diesel, low battery, low tank with no power, spike, controller offline, generator service due) are pushed to the manager's phone; once per hour per alert.
- Per-minute totals roll up into a per-day history that builds the weekly and monthly reports.

5 What the dashboard shows, and where it comes from

ON THE DASHBOARD	MEASURED BY	SHOWN IN
Source (NEPA / Gen / Inverter / None)	U2, U3 + PM1 voltage	Power panel, header pill, 24-hour source strip, report timelines
Voltage and safe band 200–250 V	PM1	Power panel gauge and sparkline; spike alert above 260 V
Load in watts, energy in kWh	PM1	Power panel, appliances list, report energy total
Battery %, V, A, time left	Inverter link	Inverter & battery panel, forecast, low-battery alert
Diesel litres, used today, run time left	Fuel sensor	Generator & diesel panel; drop and short-delivery alerts; report
Generator hours	U3 run time	Service countdown; service-due alert
Tank %, pump state	Tank sensor + R1	Water panel; pump auto start 20% / stop 95%; low-tank alert
Relay and load states	A1 relay feedback	Appliances, rooms, event log with the reason for each switch
Controller online, Wi-Fi strength	A1 report	Header pill, installer screen; offline alert after 5 s

6 Software components

COMPONENT	BUILT WITH	RESPONSIBILITIES
Firmware	C++ (Arduino core for ESP32)	Wi-Fi with auto-reconnect and automatic hub discovery; sensor drivers; 1 s report; applies relay commands; fail-safe after 10 s without the hub; status LED.
Hub	Python 3, standard library only	Device API; rules engine; event log; per-day history; alerts (ntfy / Telegram); building settings; users, PINs and roles; serves the dashboard and reports.
Dashboard	One self-contained web page	Live panels for power, inverter, diesel, water, security and rooms; switching with confirmation of the hub's result; estate view; installer setup; works on phone, tablet and wall screen; no app to install.
Reports	Generated from the hub's history	Weekly / monthly: hours per source, diesel used and cost, unexplained loss, deliveries, generator service, lowest battery; printable to PDF.

Security and access

- Controller must present a per-building device key.
- People log in with PINs; roles: owner, manager, caretaker, resident. Five wrong PINs lock a device out.
- No inbound internet access is needed; alerts go out over HTTPS.
- Every switch is logged with who or which rule did it.

When something fails

- **Wi-Fi or hub down**: relays go to safe states; dashboard shows Not live.
- **Internet down**: everything works; alerts are sent when it returns, and show on the dashboard meanwhile.
- **Sensor fault**: reading marked missing, not guessed; the hub keeps working on the rest.
- **Power cut**: hub and router on the inverter; the controller reports the outage itself.

7 Status today

PART	STATUS	NOTES
Controller ↔ hub link, relays, fail-safe, hub discovery	**Working**	Tested on an ESP32 with relay outputs
Rules, event log, history, reports, alerts to phone	**Working**	Demonstrated live
Dashboard, roles and PINs, building setup, installer screen	**Working**	Runs from the hub on any phone or laptop
PZEM power meter, AC source detection	**Next**	Standard modules; first items on the pilot build
Fuel probe, tank ultrasonic, inverter RS485	**Next**	Hub already accepts these readings; inverter register map per brand

Proposed next step: a site survey of one Max-Migold building (distribution board, changeover, generator, inverter brand, tanks), then a fixed parts list and a 4–6 week pilot with weekly reports.